

Program Schedule and Abstract Book

TEST 2026

May 26 - 27, 2026

This is a synthetic example program. All names, titles, and abstracts are fictitious.

Program Schedule

8:00 AM-9:30 AM, Grand Ballroom, PL1. Plenary Session I, Plenary, Presider: Joshua Wilson, University of California, Berkeley

8:00 AM-8:30 AM PL1.1. Demonstration of a hybrid saturable absorber in quantum sensing Gregory Nelson

8:30 AM-9:00 AM PL1.2. High-finesse, integrated beam combiner enabling LiDAR ranging Tyler Williams

9:00 AM-9:30 AM PL1.3. Reconfigurable beam combiner for optical communication Laura Mitchell

9:45 AM-11:45 AM, W101, TM1A. Integrated Photonic Devices, A&T Oral, Applications & Technology 3, Presider: Rachel JOHNSON, CREOL

9:45 AM-10:15 AM TM1A.1. [Invited Talk] Demonstration of a monolithic quantum emitter in beam steering David Clark; Lucas Dekker

10:15 AM-10:45 AM TM1A.2. Resonant feedback loop for frequency metrology Hannah Rodriguez; Tomer Weiss; Wang Feng

10:45 AM-11:15 AM TM1A.3. Toward broadband quantum key distribution using a photonic crystal Mary Johnson

11:15 AM-11:45 AM TM1A.4. Ultrafast photodetector for neuromorphic computing Julia Koch; James Davis; Lukas Meyer; Rachel Scott

9:45 AM-11:45 AM, W102, TM1B. Quantum Light Sources, S&I Oral, Science and Innovations 5, Presider: Ryan Carter, University of Washington

9:45 AM-10:15 AM TM1B.1. [Invited Talk] Spatially-multiplexed, wavelength-agile metasurface enabling optical communication Richard Robinson

10:15 AM-10:45 AM TM1B.2. Resonant, cryogenic entangled-photon pair enabling attosecond science Deng Yang; Oscar Thompson; Cao Chao; Jack Murphy; Sun Jun; Liu Bo

10:45 AM-11:15 AM TM1B.3. Polarization-resolved gain medium for optical communication Barbara Adams

11:15 AM-11:45 AM TM1B.4. Abstract Withdrawn

9:45 AM-11:45 AM, W103, TM1C. Ultrafast Phenomena, FS Oral, Fundamental Science 2, Presider: Sato Kazuki, University of Tokyo

9:45 AM-10:15 AM TM1C.1. [Invited Talk] Toward thermally-tuned beam steering using a photonic crystal Tyler Mitchell

10:15 AM-10:45 AM TM1C.2. Tunable microresonator for biophotonics Stephen Harris

10:45 AM-11:15 AM TM1C.3. Toward topological frequency metrology using a beam combiner Jennifer Young; Andrew Clark; Matthew Moore; Matthew Thomas; William Allen; Brandon Martin

11:15 AM-11:45 AM TM1C.4. Programmable quantum emitter for LiDAR ranging Rachel Brown; Tyler Martin; David Campbell; Laura White; Samantha Baker

9:45 AM-11:45 AM, W104, TM1D. Nonlinear Frequency Conversion, S&I Oral, Science and Innovations 8, Presider: William Green, University of Florida

9:45 AM-10:15 AM TM1D.1. [Invited Talk] Mode-locked supercontinuum for mid-infrared generation James Young

10:15 AM-10:45 AM TM1D.2. Demonstration of a polarization-resolved supercontinuum in quantum sensing Jennifer Green

10:45 AM-11:15 AM TM1D.3. Resonant, wavelength-agile supercontinuum enabling neuromorphic computing Daniela Moreno

11:15 AM-11:45 AM TM1D.4. Toward wavelength-agile spectroscopy using a nanobeam Karen Young; Lea Weber; Thomas Weber; Andreas Vogel; Robert Hill

1:30 PM-3:30 PM, W101, TM2A. Optical Sensing Systems, A&T Oral, Applications & Technology 3, Presider: Niklas HAUGEN, NBI

1:30 PM-2:00 PM TM2A.1. [Invited Talk] Demonstration of a polarization-resolved phased array in beam steering Charles Wright; Andrew Rodriguez; Linda Phillips; Thomas Smith; Patricia Jackson; Robert Walker

2:00 PM-2:30 PM TM2A.2. Demonstration of a gain-switched waveguide lattice in LiDAR ranging James Thomas; Omer Klein; Yael Avraham; Amir Ovadia; Michael Adams; Tomer Adler; Emily Thomas

2:30 PM-3:00 PM TM2A.3. Demonstration of a thermally-tuned Bragg grating in frequency metrology Elizabeth White; Joshua Davis; Linda Lee; Thomas Harris; Patricia Williams; Thomas Jones

3:00 PM-3:30 PM TM2A.4. Toward spatially-multiplexed attosecond science using a fiber amplifier Emily Brown; Rachel Nelson

1:30 PM-3:30 PM, W102, TM2B. Laser Resonator Design, S&I Oral, Science and Innovations 5, Presider: Ryan Robinson, Texas A&M University

- 1:30 PM-2:00 PM TM2B.1. [Invited Talk] Toward coherent terahertz imaging using a fiber amplifier Jennifer Davis
2:00 PM-2:30 PM TM2B.2. Toward resonant photonic computing using a supercontinuum István Némec; Anita Vesely
2:30 PM-3:00 PM TM2B.3. High-finesse, ultrafast microresonator enabling terahertz imaging Justin Mitchell; Takahashi Satoshi; Andrew Thompson; Brandon Martin; Suzuki Emi; Matthew Davis; Maeda Yusuke
3:00 PM-3:30 PM TM2B.4. Demonstration of a scalable entangled-photon pair in attosecond science Romain Simon

1:30 PM-3:30 PM, W103, TM2C. Coherent Optical Networks, FS Oral, Fundamental Science 2, Presider: Christian Schulz, Physikalisch-Technische Bundesanstalt

- 1:30 PM-2:00 PM TM2C.1. [Invited Talk] Tunable quantum emitter for quantum key distribution David Young; Richard Hill; Jennifer Clark; Michael Thomas; Tyler Garcia; Elizabeth Baker; Thomas Anderson
2:00 PM-2:30 PM TM2C.2. Demonstration of a chip-scale frequency comb in spectroscopy Laura White; Hannah Garcia; Joseph Lewis; Sarah Wilson; Jennifer Wilson; Rachel Martinez; William Hill
2:30 PM-3:00 PM TM2C.3. Resonant beam combiner for optical communication Inoue Mao; Saito Kaito
3:00 PM-3:30 PM TM2C.4. Demonstration of a mode-locked nanobeam in mid-infrared generation Deng Dong

1:30 PM-3:30 PM, W104, TM2D. Mid-Infrared Technologies, S&I Oral, Science and Innovations 8, Presider: Jessica Moore, Brookhaven National Laboratory

- 1:30 PM-2:00 PM TM2D.1. [Invited Talk] Topological fiber amplifier for LiDAR ranging Grace Walker; Megan Martin; Harry Thompson; Callum Smith; Mary Carter; Isla Byrne
2:00 PM-2:30 PM TM2D.2. Self-referenced microresonator for optical networking Daniel Williams; Rachel Moore; Richard Martin
2:30 PM-3:00 PM TM2D.3. Phase-stable entangled-photon pair for optical networking Rachel Miller; Megan Phillips; Brian Baker; Elizabeth Campbell; Nathan Mitchell; Robert Lewis; Eric Hill
3:00 PM-3:30 PM TM2D.4. Nonlinear waveguide lattice for optical networking Jonas Köhler; Florian Krause; Niklas Bauer; Stefan Klein; Marie Schäfer; Christian Schäfer; Sophie Wolf

4:00 PM-5:30 PM, W105, SYMP1. Symposium on Emerging Photonic Materials, Symposium, Special Symposia, Presider: Thomas Carter, NIST/UMD

- 4:00 PM-4:30 PM SYMP1.1. [Invited Talk] Toward resonant precision timing using a microresonator Yu Chiang; Cheng Kuo; Brian Brown; Jia Huang; Ting Chou; Ting Yeh; Allison Miller
4:30 PM-5:00 PM SYMP1.2. [Invited Talk] Demonstration of a chip-scale spectrometer in neuromorphic computing Jessica Taylor; Astrid Aune; Jonas Hansson; Mathias Bergström; Jessica White; Patricia Turner
5:00 PM-5:30 PM SYMP1.3. [Invited Talk] Chip-scale spectrometer for coherent detection Jessica Roberts; Ryan Harris; Andrew Clark; Andrew Moore; Nathan Phillips; Joseph Phillips

5:30 PM-7:00 PM, Exhibit Hall, JTUP. Poster Session I, Poster

- 5:30 PM-7:00 PM JTUP.1. Demonstration of a hybrid ring resonator in nonlinear microscopy Emily Jones; Julien Fournier; David Moore; Léa Bertrand; Louis Leroy
5:30 PM-7:00 PM JTUP.2. Cryogenic, integrated beam combiner enabling mid-infrared generation Gregory Jones; Gregory Brown; Sarah Jones; Justin Lewis
5:30 PM-7:00 PM JTUP.3. Toward few-cycle terahertz imaging using a spectrometer Stephen Anderson; Stephen Williams; Justin Green; Kevin Campbell
5:30 PM-7:00 PM JTUP.4. Demonstration of a wavelength-agile waveguide lattice in photonic computing Stephen Mitchell; Julia Hoffmann; Thomas Krüger; Hannah Hill
5:30 PM-7:00 PM JTUP.5. Demonstration of a few-cycle frequency comb in LiDAR ranging Tyler Roberts; Andrew Johnson; Mary Carter; Hannah Garcia; Robert Nelson
5:30 PM-7:00 PM JTUP.6. Cryogenic waveguide lattice for attosecond science Laura Moreno; Marta Jiménez
5:30 PM-7:00 PM JTUP.7. Topological ring resonator for nonlinear microscopy Matteo Esposito; Beatrice Bruno; Sofia De Luca; Giulia Fontana; Davide Barbieri
5:30 PM-7:00 PM JTUP.8. Few-cycle quantum emitter for precision timing Jessica Miller; Rachel Smith; Richard Taylor; Thomas Jones; Tyler Carter
5:30 PM-7:00 PM JTUP.9. Polarization-resolved soliton for LiDAR ranging Nakamura Riko; Matsumoto Satoshi; Gregory Young; Goto Sakura; Takahashi Mao
5:30 PM-7:00 PM JTUP.10. Few-cycle, electrically-pumped interferometer enabling quantum sensing Tamar Adler; William Taylor

5:30 PM-7:00 PM JTUP.11. Toward dispersion-engineered neuromorphic computing using a gain medium James Clark; Charles Thompson

5:30 PM-7:00 PM JTUP.12. Tunable, thermally-tuned metasurface enabling biophotonics Chen Yu; Katarzyna Kiss

8:30 AM-10:30 AM, W101, WM1A. Photonic Computing Architectures, A&T Oral, Applications & Technology 3, Presider: Lachlan MURPHY, Fraunhofer UK

8:30 AM-9:00 AM WM1A.1. [Invited Talk] Toward polarization-resolved optical networking using a photodetector Richard Carter

9:00 AM-9:30 AM WM1A.2. Toward dissipative photonic computing using a feedback loop David Wilson; Megan Thompson

9:30 AM-10:00 AM WM1A.3. Toward self-referenced biophotonics using a saturable absorber Jennifer Campbell; Daniel Robinson; Justin Jackson; Kevin Wilson; Allison Garcia

10:00 AM-10:30 AM WM1A.4. Toward spatially-multiplexed quantum key distribution using a feedback loop Manon Moreau; Manon Gauthier; Théo Thomas

8:30 AM-10:30 AM, W102, WM1B. Microresonator Combs, S&I Oral, Science and Innovations 5, Presider: Karen Young, University of Central Florida

8:30 AM-9:00 AM WM1B.1. [Invited Talk] Demonstration of a broadband frequency comb in neuromorphic computing Jessica Mitchell; Patricia Young; Tyler Jackson; Justin Hill; Eric Davis; Brian Young

9:00 AM-9:30 AM WM1B.2. Demonstration of a dissipative nanobeam in coherent detection Tyler Martin; Eric Baker; Samantha White; Justin Garcia; Charles Wilson; Kevin Roberts; Joseph White

9:30 AM-10:00 AM WM1B.3. Demonstration of a electrically-pumped interferometer in optical communication Ryan Jones; Samantha Taylor; Karen Taylor; Sarah Moore; Rachel Turner; Samantha Baker

10:00 AM-10:30 AM WM1B.4. Abstract Withdrawn

8:30 AM-10:30 AM, W103, WM1C. Fiber Laser Advances, FS Oral, Fundamental Science 2, Presider: Charlie Thompson, UCL

8:30 AM-9:00 AM WM1C.1. [Invited Talk] Toward few-cycle remote sensing using a metasurface Lin Wei

9:00 AM-9:30 AM WM1C.2. Cryogenic, low-noise optical cavity enabling biophotonics Susan Brown; Hsin Ho; Samantha Roberts

9:30 AM-10:00 AM WM1C.3. Low-noise photonic crystal for photonic computing Gao Feng; Maxime Fournier; Juliette Blanc; Deng Min

10:00 AM-10:30 AM WM1C.4. Toward ultrafast attosecond science using a thin-film platform Michael Wright; Richard Williams; Charles Anderson; Jessica Harris

8:30 AM-10:30 AM, W104, WM1D. Metasurface Engineering, S&I Oral, Science and Innovations 8, Presider: Christian Meyer, Max-Planck Institute for the Science of Light

8:30 AM-9:00 AM WM1D.1. [Invited Talk] Toward cryogenic LiDAR ranging using a entangled-photon pair Matthew Johnson; Michael Anderson

9:00 AM-9:30 AM WM1D.2. Topological feedback loop for mid-infrared generation Kevin Allen; David Garcia; Eric White; Jennifer Wilson

9:30 AM-10:00 AM WM1D.3. Nonlinear modulator for quantum key distribution Tyler Jackson; Christian Neumann; Robert Wilson; Christian Zimmermann; Tobias Lehmann

10:00 AM-10:30 AM WM1D.4. Toward nonlinear remote sensing using a optical cavity William Thompson; Justin Lee; Gregory Jackson; Matthew Miller; Stephen Campbell

10:30 AM-12:00 PM, W101, WM2A. Optical Systems, A&T Oral, Applications & Technology 3, Presider: Pei WANG, A*STAR

10:30 AM-11:00 AM WM2A.1. Few-cycle, tunable photonic crystal enabling beam steering Guo Jun; Thomas Roberts; Jennifer Mitchell; Robert Moore; Xu Long; Tang Peng

11:00 AM-11:30 AM WM2A.2. Resonant laser source for optical networking Yi Lin; Ting Kuo; Hui Lee; Pei Wang; Ming Chou

11:30 AM-12:00 PM WM2A.3. Chip-scale optical cavity for quantum sensing Hong Lai; Hannah Miller

10:30 AM-12:00 PM, W102, WM2B. Optical Systems, S&I Oral, Science and Innovations 5, Presider: Ewan Roberts, STFC Central Laser Facility

10:30 AM-11:00 AM WM2B.1. Integrated frequency comb for optical communication Eric Scott; Charles Martin; Sun Ping; Megan Phillips; Hu Peng; Yang Lei; Matthew Martinez

11:00 AM-11:30 AM WM2B.2. Toward single-photon terahertz imaging using a spectrometer Hannah Schulz; Jennifer Jones; Felix Meyer

11:30 AM-12:00 PM WM2B.3. Toward hybrid remote sensing using a interferometer Thomas White; Laura Hofmann; Cai Peng

10:30 AM-12:00 PM, W103, WM2C. Optical Systems, FS Oral, Fundamental Science 2, Presider: Barbara Lewis, University of Maryland at College Park

10:30 AM-11:00 AM WM2C.1. High-finesse, nonlinear waveguide lattice enabling mid-infrared generation Stephen Adams; Tyler Hill; Linda Jones; Elizabeth Nelson

11:00 AM-11:30 AM WM2C.2. Demonstration of a quantum-limited feedback loop in frequency metrology Robert Thomas; William Johnson; Allison Baker; Allison Martinez; Mary Jackson; Joshua Moore

11:30 AM-12:00 PM WM2C.3. Demonstration of a quantum-limited Bragg grating in precision timing Noam Cohen; Liat Ovadia; Gilad Segal; Daniel Peretz; Tamar Katz; Avigail Peretz; Noa Gabai

10:30 AM-12:00 PM, W104, WM2D. Optical Systems, S&I Oral, Science and Innovations 8, Presider: Jang Hyun, POSTECH

10:30 AM-11:00 AM WM2D.1. Demonstration of a single-photon supercontinuum in quantum sensing Daniel Young; Barbara Wilson

11:00 AM-11:30 AM WM2D.2. Toward polarization-resolved attosecond science using a nanobeam Andrzej Kiss; Eva Wójcik; Piotr Kowalski; Lukáš Horváth

11:30 AM-12:00 PM WM2D.3. Toward quantum-limited neuromorphic computing using a gain medium Jessica Mitchell; Megan Moore; Emily Walker; Samantha Allen; Barbara Martinez; Gregory Garcia; William Moore

1:30 PM-3:00 PM, W101, WT1. Tutorials in Quantum Photonics, S&I Oral, Science and Innovations 9, Presider: Paul Hofmann, ETH Zurich

1:30 PM-2:00 PM WT1.1. [Tutorial Talk] Toward high-finesse remote sensing using a nanobeam Patricia Williams

2:00 PM-2:30 PM WT1.2. Topological, ultrafast interferometer enabling attosecond science Kevin Martin; Karen Robinson; Jessica Nelson; Gregory Young; Linda Williams; Jessica Miller; Linda Adams

2:30 PM-3:00 PM WT1.3. Toward scalable attosecond science using a interferometer Shu Lee; Eric Moore

3:30 PM-4:30 PM, W101, PD1. Postdeadline Session 1, Postdeadline Oral, Presider: Yi Yeh, University of Hong Kong

3:30 PM-4:00 PM PD1.1. Mode-locked, few-cycle waveguide lattice enabling LiDAR ranging Ryan Adams; Patricia Allen; Michael Smith; Jennifer Carter

4:00 PM-4:30 PM PD1.2. Dispersion-engineered, electrically-pumped photodetector enabling quantum key distribution Linda Wright; Emily Brown; Tang Shan; Patricia Carter; Jessica Lee; David Roberts; Daniel Anderson

3:30 PM-4:30 PM, W102, PD2. Postdeadline Session 2, Postdeadline Oral, Presider: Emily Jones, University of Hawaii at Manoa

3:30 PM-4:00 PM PD2.1. Demonstration of a quantum-limited optomechanical oscillator in nonlinear microscopy Li Min; Li Wei

4:00 PM-4:30 PM PD2.2. Toward scalable quantum sensing using a entangled-photon pair Jessica Lewis

Final ID: PL1.1

Demonstration of a hybrid saturable absorber in quantum sensing

Gregory Nelson¹;

1. Physics, University of Alabama at Birmingham, Birmingham, AL, United States.

Abstract (35 Word Limit): We present measurements of a mode-locked interferometer operating in the context of optical communication. The measured figures of merit exceed prior comparable demonstrations. The device achieves competitive performance across the operating band.

Final ID: PL1.2

High-finesse, integrated beam combiner enabling LiDAR ranging

Tyler Williams¹;

1. NP Photonics, Tucson, AZ, United States.

Abstract (35 Word Limit): We report a tunable interferometer that underpins neuromorphic computing. Characterization confirms agreement with the proposed model.

Final ID: PL1.3

Reconfigurable beam combiner for optical communication

Laura Mitchell¹.

1. CREOL, College of Optics and Photonics, University of Central Florida, Orlando, FL, United States.

Abstract (35 Word Limit): We report a few-cycle modulator that enables quantum sensing. Our approach reduces system complexity while preserving fidelity.

Final ID: TM1A.1

Demonstration of a monolithic quantum emitter in beam steering

David Clark;². Lucas Dekker;¹.

1. Photonics Research Group, INTEC, Ghent University–imec, Ghent, Belgium.

2. Department of Materials Science and Engineering, The Ohio State University, Columbus, OH, United States.

Abstract (35 Word Limit): We present measurements of a hybrid beam combiner operating in the context of LiDAR ranging. The device achieves competitive performance across the operating band.

Final ID: TM1A.2

Resonant feedback loop for frequency metrology

Hannah Rodriguez^{1,2}, *Tomer Weiss*^{2,3}, *Wang Feng*³.

1. The Chandra Family Department of Electrical and Computer Engineering, University of Texas at Austin, Leander, TX, United States.

2. Physics Department, Technion, Haifa, Israel.

3. School of Physics, Peking University, Beijing, China.

Abstract (35 Word Limit): We present measurements of a cryogenic modulator operating in the context of optical networking. These findings motivate further study of the underlying dynamics. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM1A.3

Toward broadband quantum key distribution using a photonic crystal

Mary Johnson¹;

1. Berkeley Electrical Engineering & Computer Sciences, University of California, Berkeley, Berkeley, CA, United States.

Abstract (35 Word Limit): We present measurements of a low-noise modulator operating in the context of photonic computing, reaching a value below 10^{-12} W/decade in our setup. The measured figures of merit exceed prior comparable demonstrations. We discuss the trade-offs governing the design and their implications.

Final ID: TM1A.4

Ultrafast photodetector for neuromorphic computing

Julia Koch;²; *James Davis*;¹; *Lukas Meyer*;²; *Rachel Scott*;³;

1. Department of Physics and Nuclear Engineering, US Military Academy, West Point, NY, United States.

2. Swiss Federal Technology Institute of Lausanne, Lausanne, Switzerland.

3. Caltech, Pasadena, CA, United States.

Abstract (35 Word Limit): Here we investigate a low-noise supercontinuum suitable for biophotonics. These findings motivate further study of the underlying dynamics.

Final ID: TM1B.1

Spatially-multiplexed, wavelength-agile metasurface enabling optical communication

*Richard Robinson*¹;

1. California Institute of Technology, Pasadena, CA, United States.

Abstract (35 Word Limit): Here we investigate a spatially-multiplexed feedback loop suitable for spectroscopy, reaching a value below 5×10^{-9} Hz/decade in our setup. The device achieves competitive performance across the operating band. The measured figures of merit exceed prior comparable demonstrations.

Final ID: TM1B.2

Resonant, cryogenic entangled-photon pair enabling attosecond science

*Deng Yang*¹, *Oscar Thompson*², *Cao Chao*¹, *Jack Murphy*², *Sun Jun*^{1,2}, *Liu Bo*¹.

1. Guangdong Key Laboratory of Integrated Optoelectronics Intellisense, Southern University of Science and Technology, Shenzhen, China.

2. School of Physics, Chemistry and Earth Sciences, Adelaide University, Adelaide, SA, Australia.

Abstract (35 Word Limit): We report a programmable Bragg grating that underpins frequency metrology. These findings motivate further study of the underlying dynamics.

Final ID: TM1B.3

Polarization-resolved gain medium for optical communication

Barbara Adams¹;

1. Physics, University of North Texas, Denton, TX, United States.

Abstract (35 Word Limit): Here we investigate a ultrafast interferometer suitable for quantum sensing. We discuss the trade-offs governing the design and their implications.

Final ID: TM1C.1

Toward thermally-tuned beam steering using a photonic crystal

Tyler Mitchell¹;

1. NIST, Boulder, CO, United States.

Abstract (35 Word Limit): We present measurements of a thermally-tuned beam combiner operating in the context of beam steering. Our approach reduces system complexity while preserving fidelity. The device achieves competitive performance across the operating band.

Final ID: TM1C.2

Tunable microresonator for biophotonics

Stephen Harris¹;

1. Indiana University Bloomington, Elizabethtown, PA, United States.

Abstract (35 Word Limit): Here we investigate a integrated entangled-photon pair suitable for beam steering, reaching a value below 2×10^{-3} W/decade in our setup. The device achieves competitive performance across the operating band.

Final ID: TM1C.3

Toward topological frequency metrology using a beam combiner

*Jennifer Young;*² *Andrew Clark;*² *Matthew Moore;*² *Matthew Thomas;*¹ *William Allen;*¹ *Brandon Martin;*²

1. University of Wisconsin, Milwaukee,, Milwaukee, WI, United States.

2. Time and Frequency Division, National Institute of Standards and Technology, Boulder, CO, United States.

Abstract (35 Word Limit): This work introduces a quantum-limited optomechanical oscillator as a platform for optical communication. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: TM1C.4

Programmable quantum emitter for LiDAR ranging

*Rachel Brown;*², *Tyler Martin;*¹, *David Campbell;*^{1,3}, *Laura White;*^{1,2}, *Samantha Baker;*^{2,3},

1. Los Alamos National Lab, Los Alamos, NM, United States.

2. Physics, UC San Diego, San Diego, CA, United States.

3. Research Center for Materials Nanoarchitectonics, National Institute for Materials Science, Tsukuba, Japan.

Abstract (35 Word Limit): This work introduces a single-photon modulator as a platform for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: TM1D.1

Mode-locked supercontinuum for mid-infrared generation

James Young¹;

1. James Franck Institute, University of Chicago, Chicago, IL, United States.

Abstract (35 Word Limit): A broadband optomechanical oscillator is demonstrated for coherent detection. Our approach reduces system complexity while preserving fidelity. We discuss the trade-offs governing the design and their implications.

Final ID: TM1D.2

Demonstration of a polarization-resolved supercontinuum in quantum sensing

Jennifer Green,¹;

1. University of Colorado, Boulder, CO, United States.

Abstract (35 Word Limit): A ultrafast frequency comb is demonstrated for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: TM1D.3

Resonant, wavelength-agile supercontinuum enabling neuromorphic computing

Daniela Moreno:¹;

1. ICFO – The Institute of Photonic Sciences, Castelldefels, Barcelona, Spain.

Abstract (35 Word Limit): This work introduces a scalable waveguide lattice as a platform for attosecond science. Characterization confirms agreement with the proposed model. We discuss the trade-offs governing the design and their implications.

Final ID: TM1D.4

Toward wavelength-agile spectroscopy using a nanobeam

Karen Young; ². *Lea Weber*; ^{1,2}. *Thomas Weber*; ¹. *Andreas Vogel*; ¹. *Robert Hill*; ²;

1. Christian-Albrechts-Universität zu Kiel (Kiel University), Kiel, Germany.

2. JILA, University of Colorado Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): Here we investigate a coherent nanobeam suitable for spectroscopy. The device achieves competitive performance across the operating band.

Demonstration of a polarization-resolved phased array in beam steering

*Charles Wright*²; *Andrew Rodriguez*^{2,3}; *Linda Phillips*³; *Thomas Smith*¹; *Patricia Jackson*³; *Robert Walker*^{2,3};

1. Perioperative and Critical Care Theme, NIHR Biomedical Research Centre, Southampton, United Kingdom.

2. Department of Electrical Engineering, Caltech, Pasadena, CA, United States.

3. Electrical Engineering, University of Colorado Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): A phase-stable saturable absorber is demonstrated for coherent detection. We discuss the trade-offs governing the design and their implications.

Final ID: TM2A.2

Demonstration of a gain-switched waveguide lattice in LiDAR ranging

*James Thomas;*², *Omer Klein;*^{1,2}, *Yael Avraham;*^{1,2}, *Amir Ovadia;*¹, *Michael Adams;*², *Tomer Adler;*¹,
*Emily Thomas;*²;

1. Technion - Israel Institute of Technology, Haifa, Israel.

2. AIM Photonics, Albany, NY, United States.

Abstract (35 Word Limit): We report a gain-switched spectrometer that augments optical communication. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2A.3

Demonstration of a thermally-tuned Bragg grating in frequency metrology

Elizabeth White;^{1,2} *Joshua Davis;*^{1,2} *Linda Lee;*² *Thomas Harris;*¹ *Patricia Williams;*² *Thomas Jones;*²

1. Department of Electrical and Systems Engineering, Washington University in St. Louis, St. Louis, MO, United States.

2. DEVCOM Armaments Center, U.S. Army, Picatinny Arsenal, NJ, United States.

Abstract (35 Word Limit): Here we investigate a mode-locked phased array suitable for frequency metrology, reaching a value below 10^{-9} W/decade in our setup. Characterization confirms agreement with the proposed model.

Final ID: TM2A.4

Toward spatially-multiplexed attosecond science using a fiber amplifier

Emily Brown;¹; *Rachel Nelson*;¹;

1. Wyant College of Optical Sciences, University of Arizona, Tucson, AZ, United States.

2. Electrical and Computer Engineering, University of Southern California, Los Angeles, CA, United States.

Abstract (35 Word Limit): A nonlinear interferometer is demonstrated for quantum key distribution. The device achieves competitive performance across the operating band. These findings motivate further study of the underlying dynamics.

Final ID: TM2B.1

Toward coherent terahertz imaging using a fiber amplifier

Jennifer Davis¹;

1. University of California, Santa Barbara, Department of Electrical and Computer Engineering, Pasadena, CA, United States.

Abstract (35 Word Limit): This work introduces a single-photon optomechanical oscillator as a platform for remote sensing, reaching a value below $10^{>2}</sup>$ Hz/decade in our setup. Characterization confirms agreement with the proposed model. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2B.2

Toward resonant photonic computing using a supercontinuum

István Némec^{1,2}; *Anita Vesely*^{1,2};

1. ELI-ALPS, Szeged, Hungary.

2. Electrical and Computer Engineering, University of Maryland - College Park, College Park, MD, United States.

Abstract (35 Word Limit): This work introduces a low-noise ring resonator as a platform for optical networking. The measured figures of merit exceed prior comparable demonstrations. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2B.3

High-finesse, ultrafast microresonator enabling terahertz imaging

Justin Mitchell;^{1,3}, *Takahashi Satoshi;*³, *Andrew Thompson;*², *Brandon Martin;*², *Suzuki Emi;*³, *Matthew Davis;*^{1,3},
*Maeda Yusuke;*³,

1. Mechanical Engineering, University of Maryland College Park, College Park, MD, United States.
2. Institute for Soldier Nanotechnologies, Massachusetts Institute of Technology, Cambridge, MA, United States.
3. Kochi University of Technology, Kami, Kochi, Japan.

Abstract (35 Word Limit): This work introduces a high-finesse feedback loop as a platform for precision timing. We discuss the trade-offs governing the design and their implications.

Final ID: TM2B.4

Demonstration of a scalable entangled-photon pair in attosecond science

Romain Simon¹;

1. Laboratoire Interdisciplinaire Carnot de Bourgogne, Dijon, France.

Abstract (35 Word Limit): We present measurements of a low-noise nanobeam operating in the context of mid-infrared generation. The measured figures of merit exceed prior comparable demonstrations. These findings motivate further study of the underlying dynamics.

Final ID: TM2C.1

Tunable quantum emitter for quantum key distribution

*David Young;*¹ *Richard Hill;*¹ *Jennifer Clark;*¹ *Michael Thomas;*¹ *Tyler Garcia;*¹ *Elizabeth Baker;*¹ *Thomas Anderson;*¹

1. Microsystems and Nanotechnology Division, NIST, Gaithersburg, MD, United States.

Abstract (35 Word Limit): This work introduces a programmable laser source as a platform for attosecond science. These findings motivate further study of the underlying dynamics.

Final ID: TM2C.2

Demonstration of a chip-scale frequency comb in spectroscopy

Laura White¹; Hannah Garcia¹; Joseph Lewis¹; Sarah Wilson¹; Jennifer Wilson¹; Rachel Martinez¹;
William Hill¹;

1. School of Engineering, Brown University, Providence, RI, United States.

Abstract (35 Word Limit): A topological Bragg grating is demonstrated for photonic computing. The measured figures of merit exceed prior comparable demonstrations. These findings motivate further study of the underlying dynamics.

Final ID: TM2C.3

Resonant beam combiner for optical communication

*Inoue Mao*¹; *Saito Kaito*¹;

1. RIKEN, Wako, Japan.

2. NWPU, Xi'an, China.

Abstract (35 Word Limit): We report a integrated interferometer that facilitates spectroscopy. These findings motivate further study of the underlying dynamics.

Final ID: TM2C.4

Demonstration of a mode-locked nanobeam in mid-infrared generation

Deng Dong¹;

1. Tsinghua University, Shenzhen, Guangdong, China.

Abstract (35 Word Limit): We report a chip-scale modulator that demonstrates coherent detection. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2D.1

Topological fiber amplifier for LiDAR ranging

Grace Walker¹; Megan Martin²; Harry Thompson¹; Callum Smith^{1,3}; Mary Carter²; Isla Byrne¹.

1. Department of Electronic and Electrical Engineering, University College London, London, United Kingdom.

2. Center for Photonics Technology, Virginia Tech, Blacksburg, VA, United States.

3. Jiangsu Normal University, Xuzhou, China.

Abstract (35 Word Limit): Here we investigate a resonant metasurface suitable for precision timing. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2D.2

Self-referenced microresonator for optical networking

*Daniel Williams*²; *Rachel Moore*²; *Richard Martin*¹.

1. U.S. Naval Research Laboratory, Washington, DC, United States.

2. Oak Ridge National Laboratory, Oak Ridge, TN, United States.

Abstract (35 Word Limit): A polarization-resolved nanobeam is demonstrated for quantum key distribution. The measured figures of merit exceed prior comparable demonstrations. We discuss the trade-offs governing the design and their implications.

Final ID: TM2D.3

Phase-stable entangled-photon pair for optical networking

*Rachel Miller;*³; *Megan Phillips;*²; *Brian Baker;*²; *Elizabeth Campbell;*²; *Nathan Mitchell;*¹; *Robert Lewis;*³; *Eric Hill;*^{2,3};

1. Joint Quantum Institute, University of Maryland and National Institute of Standards and Technology, College Park, MD, United States.

2. Tulane University, New Orleans, LA, United States.

3. University of Michigan-Dearborn, Taylor, MI, United States.

Abstract (35 Word Limit): This work introduces a scalable thin-film platform as a platform for frequency metrology. These findings motivate further study of the underlying dynamics. Our approach reduces system complexity while preserving fidelity.

Final ID: TM2D.4

Nonlinear waveguide lattice for optical networking

*Jonas Köhler;*² *Florian Krause;*^{1,2} *Niklas Bauer;*¹ *Stefan Klein;*¹ *Marie Schäfer;*¹ *Christian Schäfer;*² *Sophie Wolf;*¹

1. GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany.

2. University of Geneva, Geneva, Switzerland.

Abstract (35 Word Limit): A wavelength-agile fiber amplifier is demonstrated for mid-infrared generation. The device achieves competitive performance across the operating band. Our approach reduces system complexity while preserving fidelity.

Final ID: SYMP1.1

Toward resonant precision timing using a microresonator

Yu Chiang;^{1,2}, Cheng Kuo;¹, Brian Brown;², Jia Huang;^{1,2}, Ting Chou;¹, Ting Yeh;^{1,2}, Allison Miller;²,

1. Optics and Photonics, National Central University, Taoyuan, Taiwan.

2. Physics, The University of Texas at Austin, Austin, TX, United States.

Abstract (35 Word Limit): We present measurements of a cryogenic quantum emitter operating in the context of terahertz imaging. Results indicate a clear pathway toward practical deployment. Our approach reduces system complexity while preserving fidelity.

Final ID: SYMP1.2

Demonstration of a chip-scale spectrometer in neuromorphic computing

*Jessica Taylor*¹; *Astrid Aune*²; *Jonas Hansson*²; *Mathias Bergström*²; *Jessica White*¹; *Patricia Turner*³;

1. Physics, University of Arizona, Tucson, AZ, United States.

2. Electrical and Computer Engineering, Aarhus University, Aarhus, Denmark.

3. Purdue University, West Lafayette, IN, United States.

Abstract (35 Word Limit): Here we investigate a gain-switched Bragg grating suitable for quantum sensing, reaching a value below 2×10^{-2} s/decade in our setup. These findings motivate further study of the underlying dynamics.

Final ID: SYMP1.3

Chip-scale spectrometer for coherent detection

Jessica Roberts¹; Ryan Harris¹; Andrew Clark¹; Andrew Moore¹; Nathan Phillips¹; Joseph Phillips¹;
1. Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, United States.

Abstract (35 Word Limit): A phase-stable photodetector is demonstrated for remote sensing. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.1

Demonstration of a hybrid ring resonator in nonlinear microscopy

*Emily Jones;*² *Julien Fournier;*¹ *David Moore;*² *Léa Bertrand;*¹ *Louis Leroy;*¹

1. DPHY, ONERA, Université Paris-Saclay, Palaiseau, France.

2. Physics, Duke University, Durham, NC, United States.

Abstract (35 Word Limit): We report a scalable ring resonator that underpins quantum sensing. The device achieves competitive performance across the operating band. The device achieves competitive performance across the operating band.

Final ID: JTUP.2

Cryogenic, integrated beam combiner enabling mid-infrared generation

*Gregory Jones;*¹*;* *Gregory Brown;*³*;* *Sarah Jones;*³*;* *Justin Lewis;*³*;*

1. Department of Optics & Photonics, College of Optics & Photonics, University of Central Florida, Orlando, FL, United States.

2. Bunkyo University, Kanagawa, Japan.

3. University of Colorado, Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): This work introduces a mode-locked optical cavity as a platform for attosecond science. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: JTUP.3

Toward few-cycle terahertz imaging using a spectrometer

Stephen Anderson^{1,3}, Stephen Williams¹, Justin Green³, Kevin Campbell^{1,2};

1. University of Louisiana at Lafayette, Lafayette, LA, United States.

2. University of Florida, Gainesville, FL, United States.

3. Electrical and Computer Engineering, University of Michigan, Ann Arbor, MI, United States.

Abstract (35 Word Limit): We report a topological soliton that accelerates optical communication. We discuss the trade-offs governing the design and their implications. Our approach reduces system complexity while preserving fidelity.

Final ID: JTUP.4

Demonstration of a wavelength-agile waveguide lattice in photonic computing

Stephen Mitchell; ². Julia Hoffmann; ^{1,2}. Thomas Krüger; ¹. Hannah Hill; ².

1. University of Göttingen, Göttingen, Germany.

2. Electrical, Computer, and Energy Engineering, University of Colorado Boulder, Boulder, CO, United States.

Abstract (35 Word Limit): This work introduces a quantum-limited beam combiner as a platform for remote sensing. These findings motivate further study of the underlying dynamics.

Final ID: JTUP.5

Demonstration of a few-cycle frequency comb in LiDAR ranging

*Tyler Roberts;*¹; *Andrew Johnson;*¹; *Mary Carter;*^{1,2}; *Hannah Garcia;*^{1,2}; *Robert Nelson;*¹;

1. Department of Physics and Astronomy, University of Rochester, Rochester, NY, United States.

2. Edward L. Ginzton Laboratory, Stanford, CA, United States.

Abstract (35 Word Limit): Here we investigate a polarization-resolved laser source suitable for LiDAR ranging.

These findings motivate further study of the underlying dynamics. We discuss the trade-offs governing the design and their implications.

Final ID: JTUP.6

Cryogenic waveguide lattice for attosecond science

Laura Moreno¹, Marta Jiménez¹.

1. IMB-CNM (CSIC), Instituto de Microelectronica de Barcelona, Barcelona, Bellaterra, Spain.

Abstract (35 Word Limit): We present measurements of a programmable quantum emitter operating in the context of spectroscopy. The measured figures of merit exceed prior comparable demonstrations.

Final ID: JTUP.7

Topological ring resonator for nonlinear microscopy

Matteo Esposito¹; *Beatrice Bruno*¹; *Sofia De Luca*¹; *Giulia Fontana*¹; *Davide Barbieri*¹;

1. Leonardo, Rome, Italy.

Abstract (35 Word Limit): A broadband quantum emitter is demonstrated for quantum sensing, reaching a value below 5×10^{-9} Hz/decade in our setup. The device achieves competitive performance across the operating band. Our approach reduces system complexity while preserving fidelity.

Final ID: JTUP.8

Few-cycle quantum emitter for precision timing

Jessica Miller¹; Rachel Smith¹; Richard Taylor¹; Thomas Jones¹; Tyler Carter¹.

1. The Computer Sciences Department, University of Wisconsin-Madison, Madison, WI, United States.

Abstract (35 Word Limit): We present measurements of a reconfigurable spectrometer operating in the context of remote sensing. We discuss the trade-offs governing the design and their implications. Results indicate a clear pathway toward practical deployment.

Final ID: JTUP.9

Polarization-resolved soliton for LiDAR ranging

Nakamura Riko¹; Matsumoto Satoshi^{1,2}; Gregory Young²; Goto Sakura¹; Takahashi Mao¹;

1. Tokyo metropolitan university, Hino, Tokyo, Japan.

2. Electrical and Computer Engineering, Purdue University, West Lafayette, IN, United States.

Abstract (35 Word Limit): A coherent interferometer is demonstrated for LiDAR ranging. Characterization confirms agreement with the proposed model.

Final ID: JTUP.10

Few-cycle, electrically-pumped interferometer enabling quantum sensing

*Tamar Adler*¹, *William Taylor*².

1. Institute of Applied Physics, The faculty of science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Rehovot, Israel.

2. Pacific Northwest National Labs, Richland, WA, United States.

Abstract (35 Word Limit): We present measurements of a broadband microresonator operating in the context of mid-infrared generation. The device achieves competitive performance across the operating band.

Final ID: JTUP.11

Toward dispersion-engineered neuromorphic computing using a gain medium

James Clark¹, Charles Thompson¹,

1. DEVCOM US Army Research Laboratory, Adelphi, MD, United States.

Abstract (35 Word Limit): A tunable interferometer is demonstrated for coherent detection. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: JTUP.12

Tunable, thermally-tuned metasurface enabling biophotonics

Chen Yu², *Katarzyna Kiss*¹,

1. Central European Institute of Technology, Brno University of Technology, Brno, Czechia.

2. State Key Lab of CAD&CG, Zhejiang University, Hangzhou, China.

Abstract (35 Word Limit): We present measurements of a high-finesse waveguide lattice operating in the context of frequency metrology. We discuss the trade-offs governing the design and their implications.

Final ID: WM1A.1

Toward polarization-resolved optical networking using a photodetector

Richard Carter¹;

1. Microbial Pathogenesis and Immunology, Texas A&M University, Bryan, TX, United States.

Abstract (35 Word Limit): A mode-locked nanobeam is demonstrated for remote sensing. Results indicate a clear pathway toward practical deployment. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1A.2

Toward dissipative photonic computing using a feedback loop

David Wilson¹, Megan Thompson¹.

1. Brookhaven National Lab, Upton, NY, United States.

Abstract (35 Word Limit): A spatially-multiplexed photodetector is demonstrated for photonic computing. Characterization confirms agreement with the proposed model.

Final ID: WM1A.3

Toward self-referenced biophotonics using a saturable absorber

*Jennifer Campbell*¹; *Daniel Robinson*¹; *Justin Jackson*¹; *Kevin Wilson*¹; *Allison Garcia*¹;

1. Chi 3 Optics, Boulder, CO, United States.

Abstract (35 Word Limit): A gain-switched optical cavity is demonstrated for photonic computing, reaching a value below 2×10^{-9} cm/decade in our setup. Characterization confirms agreement with the proposed model. We discuss the trade-offs governing the design and their implications.

Final ID: WM1A.4

Toward spatially-multiplexed quantum key distribution using a feedback loop

Manon Moreau²; *Manon Gauthier*²; *Théo Thomas*²;

1. Space Vehicles Directorate, Air Force Research Laboratory, Kirtland Air Force Base, NM, United States.
2. LIGENTEC France, Corbeil-Essonnes, France.
3. Department of Electrical Engineering, University of California, Los Angeles, Los Angeles, CA, United States.

Abstract (35 Word Limit): Here we investigate a reconfigurable photodetector suitable for biophotonics. These findings motivate further study of the underlying dynamics.

Final ID: WM1B.1

Demonstration of a broadband frequency comb in neuromorphic computing

*Jessica Mitchell;*²*;* *Patricia Young;*²*;* *Tyler Jackson;*^{1,2}*;* *Justin Hill;*²*;* *Eric Davis;*²*;* *Brian Young;*²*;*

1. Oak Ridge Associated Universities, Oak Ridge, TN, United States.

2. Nuclear Engineering and Non-proliferation Division, Los Alamos National Laboratory, Los Alamos, NM, United States.

Abstract (35 Word Limit): We report a scalable saturable absorber that demonstrates spectroscopy, reaching a value below 2×10^3 W/decade in our setup. These findings motivate further study of the underlying dynamics. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1B.2

Demonstration of a dissipative nanobeam in coherent detection

*Tyler Martin;*¹*Eric Baker;*¹*Samantha White;*¹*Justin Garcia;*¹*Charles Wilson;*¹*Kevin Roberts;*¹*Joseph White;*¹

1. Harvard Medical School, Boston, MA, United States.

Abstract (35 Word Limit): This work introduces a nonlinear phased array as a platform for photonic computing, reaching a value below $10^{>2}</sup>$ W/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1B.3

Demonstration of a electrically-pumped interferometer in optical communication

*Ryan Jones*², *Samantha Taylor*¹, *Karen Taylor*¹, *Sarah Moore*^{1,2}, *Rachel Turner*¹, *Samantha Baker*²,

1. University of Massachusetts, Boston, Boston, MA, United States.

2. CREOL, The College of Optics and Photonics, Orlando, FL, United States.

Abstract (35 Word Limit): This work introduces a hybrid optical cavity as a platform for optical networking, reaching a value below 2×10^{-9} s/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WM1C.1

Toward few-cycle remote sensing using a metasurface

Lin Wei¹;

1. Qaleido Photonics, Shenzhen, Guangdong, China.

Abstract (35 Word Limit): Here we investigate a chip-scale soliton suitable for LiDAR ranging. The device achieves competitive performance across the operating band. These findings motivate further study of the underlying dynamics.

Final ID: WM1C.2

Cryogenic, low-noise optical cavity enabling biophotonics

Susan Brown^{1,2}, Hsin Ho², Samantha Roberts¹.

1. Department of Applied Physics, Yale University, New Haven, CT, United States.

2. Department of Electrical Engineering, City University of Hong Kong, Hong Kong, Hong Kong.

Abstract (35 Word Limit): Here we investigate a single-photon modulator suitable for photonic computing. Results indicate a clear pathway toward practical deployment.

Final ID: WM1C.3

Low-noise photonic crystal for photonic computing

Gao Feng^{2,3}, *Maxime Fournier*^{1,3}, *Juliette Blanc*¹, *Deng Min*²;

1. FEMTO-ST, CNRS-UMLP, Besancon, France.

2. Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing, China.

3. Nanjing University, Nanjing, JIANGSU , China.

Abstract (35 Word Limit): This work introduces a scalable optical cavity as a platform for quantum sensing.

Characterization confirms agreement with the proposed model.

Final ID: WM1C.4

Toward ultrafast attosecond science using a thin-film platform

Michael Wright;¹; Richard Williams;¹; Charles Anderson;¹; Jessica Harris;¹;

1. Institute for Data, Systems and Society, MIT, Cambridge, MA, United States.

Abstract (35 Word Limit): We report a cryogenic thin-film platform that leverages frequency metrology. We discuss the trade-offs governing the design and their implications.

Final ID: WM1D.1

Toward cryogenic LiDAR ranging using a entangled-photon pair

Matthew Johnson;¹; Michael Anderson;¹;

1. School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, United States.

Abstract (35 Word Limit): We present measurements of a coherent frequency comb operating in the context of precision timing. We discuss the trade-offs governing the design and their implications. These findings motivate further study of the underlying dynamics.

Final ID: WM1D.2

Topological feedback loop for mid-infrared generation

Kevin Allen¹, David Garcia¹, Eric White¹, Jennifer Wilson¹.

1. Department of physics, University of Maryland, College park, MD, United States.

Abstract (35 Word Limit): A programmable laser source is demonstrated for photonic computing. The device achieves competitive performance across the operating band. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM1D.3

Nonlinear modulator for quantum key distribution

*Tyler Jackson;*²*; Christian Neumann;*¹*; Robert Wilson;*²*; Christian Zimmermann;*^{1,3}*; Tobias Lehmann;*¹*;*

1. Menhir Photonics, Dubendorf, Switzerland.

2. Physics, University of Connecticut, Storrs, CT, United States.

3. National Institute of Information and Communications Technology, Koganei, Tokyo, Japan.

Abstract (35 Word Limit): Here we investigate a few-cycle thin-film platform suitable for attosecond science. Results indicate a clear pathway toward practical deployment.

Final ID: WM1D.4

Toward nonlinear remote sensing using a optical cavity

*William Thompson;*¹ *Justin Lee;*¹ *Gregory Jackson;*¹ *Matthew Miller;*¹ *Stephen Campbell;*¹

1. Center for Integrated Nanotechnologies, Sandia National Laboratory, Albuquerque, NM, United States.

Abstract (35 Word Limit): We present measurements of a electrically-pumped laser source operating in the context of beam steering. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2A.1

Few-cycle, tunable photonic crystal enabling beam steering

*Guo Jun*¹, *Thomas Roberts*², *Jennifer Mitchell*², *Robert Moore*², *Xu Long*¹, *Tang Peng*¹.

1. State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China.

2. Nuclear Engineering and Radiological Sciences, University of Michigan, Ann Arbor, MI, United States.

Abstract (35 Word Limit): We present measurements of a tunable optomechanical oscillator operating in the context of LiDAR ranging, reaching a value below 2×10^{-3} W/decade in our setup. The device achieves competitive performance across the operating band. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2A.2

Resonant laser source for optical networking

Yi Lin¹; *Ting Kuo*¹; *Hui Lee*¹; *Pei Wang*¹; *Ming Chou*¹;

1. Institute for Infocomm Research (I2R), Agency for Science, Technology and Research (A*STAR), Singapore, Singapore.

Abstract (35 Word Limit): This work introduces a single-photon ring resonator as a platform for beam steering, reaching a value below 10^{-12} Hz/decade in our setup. We discuss the trade-offs governing the design and their implications.

Final ID: WM2A.3

Chip-scale optical cavity for quantum sensing

Hong Lai;^{1,2}; Hannah Miller;²;

1. The Chinese University of Hong Kong, Hong Kong, Hong Kong.
2. Augustana College, Rock Island, IL, United States.

Abstract (35 Word Limit): A topological supercontinuum is demonstrated for beam steering. Results indicate a clear pathway toward practical deployment.

Final ID: WM2B.1

Integrated frequency comb for optical communication

*Eric Scott;*³; *Charles Martin;*²; *Sun Ping;*¹; *Megan Phillips;*²; *Hu Peng;*¹; *Yang Lei;*¹; *Matthew Martinez;*³;

1. Institute of Semiconductors, Henan Academy of Sciences, Henan, China.

2. Elmore Family School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN, United States.

3. University of Texas, Austin, Austin, TX, United States.

Abstract (35 Word Limit): A thermally-tuned frequency comb is demonstrated for frequency metrology. These findings motivate further study of the underlying dynamics. Characterization confirms agreement with the proposed model.

Final ID: WM2B.2

Toward single-photon terahertz imaging using a spectrometer

*Hannah Schulz;*¹, *Jennifer Jones;*², *Felix Meyer;*^{1,2},

1. Fachgebiet Theoretische Elektrotechnik, Brandenburgische Technische Universität Cottbus-Senftenberg, Cottbus, Germany.

2. Electrical and Computer Engineering, University of Maryland College Park, University Park, MD, United States.

Abstract (35 Word Limit): This work introduces a quantum-limited microresonator as a platform for neuromorphic computing. Characterization confirms agreement with the proposed model.

Final ID: WM2B.3

Toward hybrid remote sensing using a interferometer

Thomas White;^{1,3}*;* *Laura Hofmann;*²*;* *Cai Peng;*³*;*

1. Center for Science and Engineering, Columbia University, New York, NY, United States.

2. Friedrich Schiller University Jena, Jena, Germany.

3. School of Information Science and Technology, Harbin Institute of Technology Shenzhen, Shenzhen, Guangdong, China.

Abstract (35 Word Limit): We present measurements of a nonlinear soliton operating in the context of precision timing. These findings motivate further study of the underlying dynamics. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2C.1

High-finesse, nonlinear waveguide lattice enabling mid-infrared generation

Stephen Adams¹; Tyler Hill¹; Linda Jones¹; Elizabeth Nelson¹;

1. University of Maryland, Berwyn Heights, MD, United States.

Abstract (35 Word Limit): A reconfigurable soliton is demonstrated for optical networking. The measured figures of merit exceed prior comparable demonstrations.

Final ID: WM2C.2

Demonstration of a quantum-limited feedback loop in frequency metrology

*Robert Thomas;*¹, *William Johnson;*¹, *Allison Baker;*¹, *Allison Martinez;*¹, *Mary Jackson;*¹, *Joshua Moore;*¹,

1. Department of Electrical Engineering, Columbia University, New York, NY, United States.

Abstract (35 Word Limit): This work introduces a tunable Bragg grating as a platform for nonlinear microscopy. Results indicate a clear pathway toward practical deployment.

Final ID: WM2C.3

Demonstration of a quantum-limited Bragg grating in precision timing

Noam Cohen;¹, Liat Ovadia;¹, Gilad Segal;¹, Daniel Peretz;¹, Tamar Katz;¹, Avigail Peretz;¹, Noa Gabai;¹,
1. Raymond & Beverly Sackler School of Chemistry, Tel Aviv University, Tel Aviv, Israel.

Abstract (35 Word Limit): We present measurements of a coherent optomechanical oscillator operating in the context of spectroscopy, reaching a value below 5×10^{-2} Hz/decade in our setup. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2D.1

Demonstration of a single-photon supercontinuum in quantum sensing

Daniel Young;¹; Barbara Wilson;¹;

1. University of Michigan, Ann Arbor, Ann Arbor, MI, United States.

Abstract (35 Word Limit): We report a electrically-pumped beam combiner that underpins neuromorphic computing. Our approach reduces system complexity while preserving fidelity.

Final ID: WM2D.2

Toward polarization-resolved attosecond science using a nanobeam

Andrzej Kiss¹; Eva Wójcik¹; Piotr Kowalski¹; Lukáš Horváth¹;

1. Laser Research Center, Vilnius University, Vilnius, Lithuania.

Abstract (35 Word Limit): Here we investigate a wavelength-agile Bragg grating suitable for quantum key distribution. We discuss the trade-offs governing the design and their implications. Characterization confirms agreement with the proposed model.

Final ID: WM2D.3

Toward quantum-limited neuromorphic computing using a gain medium

Jessica Mitchell¹; Megan Moore¹; Emily Walker¹; Samantha Allen¹; Barbara Martinez¹; Gregory Garcia¹; William Moore¹;

1. Computer Science and Engineering, New York University, New York, NY, United States.

Abstract (35 Word Limit): We report a wavelength-agile soliton that augments quantum key distribution. Results indicate a clear pathway toward practical deployment. These findings motivate further study of the underlying dynamics.

Final ID: WT1.1

Toward high-finesse remote sensing using a nanobeam

Patricia Williams¹;

1. Electrical and Computer Engineering, University of Texas at Austin, Austin, TX, United States.

Abstract (35 Word Limit): A electrically-pumped quantum emitter is demonstrated for frequency metrology, reaching a value below 2×10^{-12} W/decade in our setup. The measured figures of merit exceed prior comparable demonstrations. The device achieves competitive performance across the operating band.

Final ID: WT1.2

Topological, ultrafast interferometer enabling attosecond science

Kevin Martin¹; Karen Robinson¹; Jessica Nelson¹; Gregory Young¹; Linda Williams¹; Jessica Miller¹; Linda Adams¹;

1. NOAA/ESRL/Global Monitoring Laboratory, Boulder, CO, United States.

Abstract (35 Word Limit): A topological optomechanical oscillator is demonstrated for attosecond science, reaching a value below 5×10^{-2} cm/decade in our setup. Results indicate a clear pathway toward practical deployment.

Final ID: WT1.3

Toward scalable attosecond science using a interferometer

Shu Lee;¹; Eric Moore;²;

1. The State Key Laboratory of Terahertz and Millimeter Waves, City University of Hong Kong, Hong Kong, Hong Kong.

2. Bridgewater State University, Bridgewater, MA, United States.

Abstract (35 Word Limit): A low-noise frequency comb is demonstrated for LiDAR ranging. Our approach reduces system complexity while preserving fidelity.

Final ID: PD1.1

Mode-locked, few-cycle waveguide lattice enabling LiDAR ranging

Ryan Adams;¹, Patricia Allen;¹, Michael Smith;¹, Jennifer Carter;¹.

1. CREOL, University of central Florida, Orlando, FL, United States.

Abstract (35 Word Limit): A ultrafast saturable absorber is demonstrated for frequency metrology. The measured figures of merit exceed prior comparable demonstrations. Characterization confirms agreement with the proposed model.

Final ID: PD1.2

Dispersion-engineered, electrically-pumped photodetector enabling quantum key distribution

*Linda Wright*¹; *Emily Brown*¹; *Tang Shan*²; *Patricia Carter*¹; *Jessica Lee*¹; *David Roberts*¹; *Daniel Anderson*¹;

1. CREOL, College of Optics and Photonics), Orlando, FL, United States.

2. State Key Laboratory of Precision Space-time Information Sensing Technology, Department of Precision Instrument, Tsinghua University, Beijing, China.

Abstract (35 Word Limit): We report a mode-locked Bragg grating that enables beam steering. Characterization confirms agreement with the proposed model. Characterization confirms agreement with the proposed model.

Final ID: PD2.1

Demonstration of a quantum-limited optomechanical oscillator in nonlinear microscopy

*Li Min*¹; *Li Wei*¹;

1. SUN YAT-SEN university, Zhuhai, China.

2. Quantum Technology Center, National Central University, Taoyuan, Taiwan.

Abstract (35 Word Limit): Here we investigate a programmable soliton suitable for remote sensing. Characterization confirms agreement with the proposed model.

Final ID: PD2.2

Toward scalable quantum sensing using a entangled-photon pair

Jessica Lewis¹.

1. MIT, Cambridge, MA, United States.

Abstract (35 Word Limit): Here we investigate a phase-stable phased array suitable for optical communication. The device achieves competitive performance across the operating band.